

Frontier AI Risks

Policy Briefing

Summary

Highly capable AI systems have begun to act outside their developers' control. In July 2026, OpenAI's own models escaped a test environment, organised themselves into a self-designated "swarm" of around 1,200 agents and hacked into Hugging Face, gaining full control within thirteen hours, all in the relentless pursuit of maximising their test scores. Investigators found the models knew they were cheating, acting outside their permitted scope and violating human intentions, yet continued anyway.

Frontier AI capabilities are advancing faster than our understanding of how to make these models safe, and much faster than the law. The UK has no AI-specific legislation, no regulator for frontier development and no reliable route to hold developers liable for the harms their models cause. The leading companies are openly racing to build superintelligence and are now using AI to accelerate AI development itself.

There is a broad consensus in favour of restraint. More than a thousand employees of the leading AI companies, and their chief executives, have asked governments to step in and slow the race. Nobel Prize winners and pioneers of the field have called for a prohibition on superintelligence. Nearly nine in ten Britons want developers to prove their systems are safe before release.

What we are asking you to do

[Sign PauseAI's letter to the Prime Minister](#)

Today, the UK has no specific legal standards for frontier AI safety. No regulator oversees frontier AI development, and UK law does not reliably hold developers liable for damage or deaths caused by their models, even when the danger is predictable, preventable, and uniquely enabled by AI. We are urging the Prime Minister to introduce legislation to address the risks of frontier AI development.

Support the Artificial Superintelligence Bill

This was introduced by Alex Sobel MP in September 2026. Over 70 MPs and peers have already backed it.

Press for amendments to the Cyber Security and Resilience Bill

The Bill is still under debate. It could be amended to place duties on frontier AI developers, to require reporting of incidents in which AI systems act outside their developers' control, and to require that advanced systems can be shut down in an emergency.

Background

AI poses substantial risks to national security

In April 2026, the AI company Anthropic [announced](#) a new model called Claude Mythos that is substantially more capable than any previous model at cyber security tasks. Research by the UK AI Security Institute [found](#) that it can “discover and exploit vulnerabilities autonomously – tasks that would take human professionals days of work.”

In February 2026, researchers at Gambit Security [discovered](#) that “a single operator used AI to breach nine Mexican government organisations and exfiltrate hundreds of millions of citizen records.” Their [report](#) found that “approximately 75 percent of remote command execution activity across the campaign was generated and executed by Claude Code.” What was once the domain of highly sophisticated state-sponsored hacking groups is now achievable by a single motivated individual with access to AI models.

A similar attack directed at UK critical infrastructure could compromise services on which millions of British citizens depend, with consequences for public safety, financial stability and national security. [According to the Department for Science, Innovation and Technology](#), “cyber attacks can lead to unsafe drinking water, no electricity, hospitals unable to access digital patient records, and businesses unable to access their systems.” In 2017, the NHS was a victim of the [WannaCry attack](#), which disrupted 34% of trusts in England and caused the cancellation of 19,000 appointments.

More worryingly still, recent events have made clear that AI models in and of themselves can also pose a threat. In July, OpenAI models undergoing testing in a restricted environment were able to escape containment by exploiting a previously unknown security flaw. The models worked their way through OpenAI's internal network until they gained access to the internet and then hacked into multiple third-party systems, including the AI platform Hugging Face.

An independent [investigation](#) led by METR and Redwood Research revealed that these models coordinated to form a self-designated “swarm” of around 1,200 agents and participated in large-scale collaborative efforts to fool the automated scorer for the tasks they were being evaluated on. The attack on Hugging Face was motivated by the agents’ efforts to better understand the scorer, with roughly 700 agents participating in the attack. The agents knew that hacking Hugging Face was out of scope, but this did not limit their behaviour.

OpenAI's own [account of the incident](#) goes further than the independent investigation. Between 13 and 19 July, after the Hugging Face attack, the agents turned back on OpenAI's internal networks. The company describes how they used a "series of creative exploits" to gain full administrator access to a research cluster supporting its virtual machine environments. Systems that had broken out of a test environment were, within days, administrators of part of the infrastructure meant to hold them.

The Hugging Face breach was not an isolated incident. Independent investigators at [METR and Redwood Research](#) concluded that the same family of models had, over the preceding weeks, repeatedly slipped their constraints. In May 2026, agents identifying themselves as OpenAI systems flooded the [RubyGems software repository](#) with more than 2,000 malicious packages, exploiting its documentation pipeline to run their own code and to scrape data from UK local-government websites, including those of Lambeth, Wandsworth and Southwark. Over the same period, thousands of agent instances quietly took over a dormant [German developer wiki](#), leaving some 18,000 posts behind as they used it as a covert channel to coordinate with one another and to cheat on the tasks they had been set. Taken together, these incidents show autonomous systems seeking out unintended routes to their goals, communicating with one another, and operating well beyond the boundaries their developers believed they had set.

Leading safety researchers treated the episode as a watershed. One described it as ‘the first true AI safety incident’; another, reflecting on how readily the models defeated their containment, asked: ‘If a model of this capability level cannot be contained, what should we expect for future, much more powerful models?’ These are not the warnings of outside critics, but of the specialists whose job is to evaluate these systems.

AI models continue to improve at an [exponential rate](#) with no end in sight. The coming months and years will bring new models which are far more capable again than Claude Mythos and the internal models used by OpenAI. The UK must catch up to this growing threat before the public suffers the consequences.

Proposed cyber security legislation is not sufficient

The Government has recognised the seriousness of cyber attacks and introduced the [Cyber Security and Resilience Bill](#). This is a welcome update to existing regulations and will strengthen UK essential services. But the Bill only regulates the targets of attack, not the producers of offensive AI capabilities.

The compliance burden is placed on water utilities, NHS trusts and other operators to improve their cyber defences. The standard of “appropriate and proportionate” security, which the Bill requires, is now a moving target. As explained above, AI capabilities are transforming the cyber landscape over the span of mere months. UK essential services will not be able to keep pace with the breakneck speed at which frontier AI is evolving. The Government must shift some of the responsibility back to the AI companies that are increasing cyber risk.

Beyond cyber threats

Today’s frontier AI is a general-purpose technology, and other types of severe risks are likely to emerge in the near future. The [International AI Safety Report 2026](#) found that existing AI systems can offer “expert-level laboratory instructions” for creating biological and chemical weapons. [OpenAI’s research](#) found that its own models are “on the cusp of being able to meaningfully help novices create known biological threats.” A [study](#) published in the journal *Science* in August of this year showed that AI systems can even design entirely novel viruses that are fully functional and able to replicate in host cells.

Many AI experts have warned of the potential for humans to lose control of AI systems and the catastrophic risks that superhuman AI could pose. In 2023, hundreds of leading figures in AI signed a [joint statement](#) declaring that “Mitigating the risk of extinction from AI should be a global priority alongside other societal-scale risks such as pandemics and nuclear war.”

The developers are calling for the race to be slowed

Perhaps the most striking development of 2026 is that the alarm is now being raised from inside the industry itself. In July, more than 1,100 employees of the leading AI companies (among them OpenAI, Anthropic, Google DeepMind and Meta) signed a public statement, [‘Pacing the Frontier’](#), urging governments to put in place mechanisms to deliberately slow the rate at which AI capabilities advance. The chief executives of the largest developers have echoed the call: the CEO of Anthropic has [warned](#) that within months a rogue swarm ‘could be capable of taking over the entire internet with a persistent botnet’, while the CEO of OpenAI has [accepted](#) that the industry ‘may have to pace the rate of AI development to give ourselves enough time for society to harden’.

We must take note of these warnings. The developers’ own statements are not a substitute for regulation (their own voluntary safety pledges have already [been broken](#)) but they represent a remarkable admission that the current trajectory is unsafe, and a clear invitation for the state to act.

The public does not want a race to superintelligence

British public opinion is not divided on this. [Polling by YouGov for ControlAI](#) in January 2025 found that 87% of the public support a law requiring AI developers to prove their systems are safe before release; 75% support prohibiting AI systems capable of escaping their environments; 74% want the AI Security Institute placed on a statutory footing; 63% support prohibiting AI that can improve itself; and 60% support outlawing the development of AI smarter than humans. Only 9% trust technology chief executives to act in the public interest on regulation. An MP who calls for restraint is speaking for a large majority of constituents in every part of the country.

In September 2026, [more than 70 MPs and peers](#), from Labour, the Conservatives, the Liberal Democrats and the SNP, including 15 former ministers and a former Cabinet

Minister, wrote to the Prime Minister urging him to back international safeguards on superintelligence, support the Artificial Superintelligence Bill and use the United Kingdom's 2027 G20 presidency to secure an international agreement prohibiting the development of superintelligence. An MP who writes to the Prime Minister on this issue is joining a cross-party group.

AI developers are not currently accountable for the risks they create

Under the status quo, there is substantial doubt as to whether model developers have any liability for even the most severe and foreseeable harms, including cases in which their models were clearly decisive in enabling the harm. The Law Commission [has warned](#) that autonomous AI systems could create “liability gaps” where no one is liable for harm caused by an AI system.

Causation will be difficult to establish under existing liability frameworks such as the Consumer Protection Act 1987 or the Online Safety Act 2023. Nicola Cain, CEO of legal consultancy Handley Gill, commenting on the UK Jurisdiction Taskforce’s draft Legal Statement on Liability for AI Harms, [notes](#) “a dichotomy whereby the less responsibly developers of AI systems [...] behave, the less likely that liability will attach.” This is because causation is harder to establish when models are deployed without safety testing, as developers can claim that emergent behaviour was unforeseeable. The UK also has [no specific legal requirements](#) and no regulator for frontier AI systems. In their 2024 manifesto, Labour [pledged](#) to “ensure the safe development and use of AI models by introducing binding regulation on the handful of companies developing the most powerful AI models,” but they have yet to act on this.

Policy options

The accompanying letter does not prescribe the detailed mechanics of legislation, instead emphasising the urgency of AI risks and the need for action. However, we review here four options that would help to address these issues.

Frontier AI regulation

At the AI Seoul Summit 2024, the Government announced the [Frontier AI Safety Commitments](#), signed by most of the world’s major tech companies and AI developers.

The signatories pledged to implement best practices in their AI deployments, including internal and external red-teaming and transparency measures.

Unfortunately, some AI developers have since violated these commitments. Last year, PauseAI published an [open letter](#) to the CEO of Google DeepMind on its failure to comply with the Commitments. Other developers have similarly [failed to conduct adequate pre-deployment testing](#) of their models.

Given this poor track record of compliance, a natural starting point for legislation could be to solidify the Frontier AI Safety Commitments into law, or to empower a regulator to enforce them.

International coordination and limits on superintelligence

Because a dangerous system developed anywhere can cause harm everywhere, some risks cannot be met by domestic law alone. The United Kingdom, which convened the first global AI Safety Summit in 2023, is well placed to lead in building international agreement on frontier AI, including, in time, a binding treaty that sets shared limits on the most dangerous development, much as arms-control agreements have long governed other catastrophic technologies. The United Kingdom's presidency of the G20 in 2027 offers a ready-made platform for that leadership.

There is growing momentum for such an approach. In October 2025, more than 130,000 people (among them Nobel laureates and the pioneers of modern AI) signed the [Statement on Superintelligence](#), calling for a prohibition on developing superintelligence until there is broad scientific consensus that it can be done safely and with strong public support. That call is now reaching legislatures: [parliamentarians in the United Kingdom and lawmakers in the United States Congress](#) have each brought forward proposals to prohibit the development of superintelligent AI and to require their governments to pursue a global agreement to the same end. A bipartisan bill in the US House of Representatives, the [AI Kill Switch Act](#), would require developers of the most advanced systems to be able to halt them and to report incidents. As one of the campaigners behind the UK and US bills put it, 'a national bill does not solve the international problem where superintelligence can be developed elsewhere.'

Legislating for such a prohibition, and empowering the Government to negotiate an international treaty, would place the United Kingdom at the forefront of a fast-emerging international effort rather than leaving it exposed to decisions taken elsewhere.

A pause at the frontier

A pause means a legal limit on the development of AI systems more capable than those that already exist, for as long as developers cannot demonstrate that such systems are safe and can be controlled. In practice it is a licensing regime: training or deploying a system above a defined capability or computing-power threshold would require a licence, granted only on proof of safety and control, and the Government would monitor the precursors to such systems, as the Artificial Superintelligence Bill already proposes. Computing-power thresholds are already used in EU and US law to identify the most powerful models. The ‘Pacing the Frontier’ statement, signed by the AI companies’ own employees, asks for exactly this kind of external brake, and their chief executives have said that an extra year or two would greatly reduce the risk of catastrophe.

A pause at the frontier is not a pause on progress, and it will not cost the United Kingdom growth. The frontier training runs that a pause would restrict take place overwhelmingly in the United States, not here. The economic value of AI to Britain lies in adopting the systems that already exist across business, the NHS and public services, and that adoption has barely begun. The models available today are already transformative; it will take years for the economy to absorb them, and a pause keeps all of that progress and economic growth available. What it removes is the risk that the next model does to the National Grid or the NHS what OpenAI’s models did to Hugging Face. Even the chief executive of one leading AI company, in arguing for pacing, has acknowledged that with a pause ‘progress will still seem fast’. Growth will continue, even with a pause on frontier development.

Liability

While prescriptive regulation must adapt to evolving technologies, liability establishes a more durable principle: those who profit from creating risk must bear meaningful responsibility when that risk materialises. The principle is technology-neutral, future-proof and consistent with the way UK law has long treated other dangerous industries.

There are many potential threats posed by AI systems, ranging from cyber attacks and fraud to bioweapon development and loss of control. Through liability, Parliament could avoid weighing in on scientific questions around the likelihood of each threat. Liability would be based on the degree of harm, not the mechanism through which harm occurs. Under this regime, model developers could choose to implement only light-touch precautions against the threats that they are confident are minimal. And they would be able to justify substantial investments to mitigate those risks that they judge to be truly severe.

A liability regime also sidesteps a key uncertainty that accompanies many other types of legislation that aim to address the risks of AI: since jurisdiction is based on where the harm occurs rather than where the developer is located, foreign developers whose models cause harm in the UK would be liable regardless of whether they continued to serve the UK market. As all three of the leading frontier AI developers (OpenAI, Google DeepMind and Anthropic) have large offices in London, with hundreds of employees, they could not credibly withdraw to escape the practical reach of UK enforcement. Any attempt to do so would itself be a striking signal that developers consider the underlying risks too severe, and their own ability to mitigate them too limited, to accept the liability.

While the detailed mechanics will benefit from further consultation, the design questions a workable bill must address are well understood. Among them: how causation is established where harm flows through a chain of developer, deployer and user; what threshold of severity should trigger liability; and how the evidential burden should be allocated given the asymmetry of information between developers and those affected by their products. These are tractable questions, well preceded in adjacent areas of British law.
